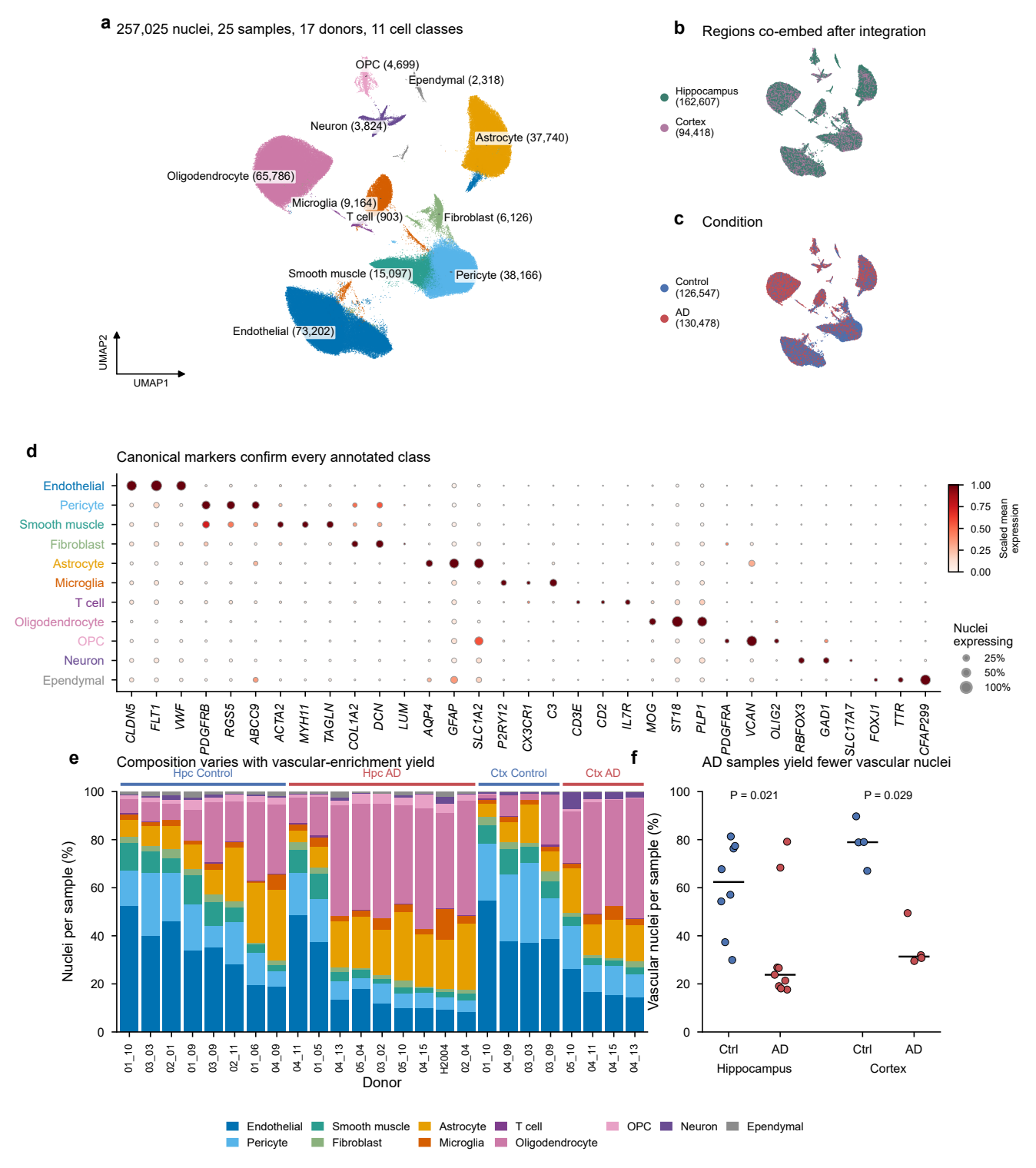

a
257,025 nuclei, 25 samples, 17 donors, 11 cell classes
OPC (4,699)
Ependymal (2,318)
Neuron (3,824)
Astrocyte (37,740)
Oligodendrocyte (65,786)
Microglia (9,164)
T cell (903)
Fibroblast (6,126)
Smooth muscle (15,097)
Pericyte (38,166)
Endothelial (73,202)
UMAP2
UMAP1
b
Regions co-embed after integration
Hippocampus (162,607)
Cortex (94,418)
c
Condition
Control (126,547)
AD (130,478)
d
Canonical markers confirm every annotated class
Endothelial
Pericyte
Smooth muscle
Fibroblast
Astrocyte
Microglia
T cell
Oligodendrocyte
OPC
Neuron
Ependymal
CLDN5
FLT1
VWF
PDGFRB
RGS5
ABCC9
ACTA2
MYH11
TAGLN
COL1A2
DCN
LUM
AQP4
GFAP
SLC1A2
P2RY12
CX3CR1
C3
CD3E
CD2
IL7R
MOG
ST18
PLP1
PDGFRA
VCAN
OLIG2
RBFOX3
GAD1
SLC17A7
FOXJ1
TTR
CFAP299
1.00
0.75
0.50
0.25
0.00
Scaled mean expression
Nuclei expressing
25%
50%
100%
e
Composition varies with vascular-enrichment yield
Hpc Control
Hpc AD
Ctx Control
Ctx AD
Nuclei per sample (%)
Donor
01_10
03_03
02_01
01_09
03_09
02_11
01_06
04_09
04_11
01_05
04_13
05_04
03_02
05_10
04_15
H2004
02_04
01_10
04_09
03_03
03_09
05_10
04_11
04_15
04_13
f
AD samples yield fewer vascular nuclei
P = 0.021
P = 0.029
Vascular nuclei per sample (%)
Ctrl
AD
Hippocampus
Ctrl
AD
Cortex
Endothelial
Pericyte
Smooth muscle
Fibroblast
Astrocyte
Microglia
T cell
Oligodendrocyte
OPC
Neuron
Ependymal